# Using Net Present Value to Prioritize State Information Technology Projects

A practical guide for comparing state IT investments when funding and resources are limited.

This guide answers two related questions: How should Net Present Value (NPV) be calculated for state IT project analysis, and how should agencies compare projects consistently when only one or a few can be funded?

Applicability. This guide does not determine whether a Schedule IV-B is required. Agencies must follow the current annual Legislative Budget Request (LBR) instructions. For the current budget cycle, a Schedule IV-B is generally required for an IT funding request with a total estimated project cost of $1 million or more, subject to the exceptions listed in those instructions. Section 216.023(4)(a)10, Florida Statutes, imposes additional governance/policy information requirements for IT projects exceeding $10 million.

## Calculate Net Present Value Consistently

Illustrative NPV concept: $NPV = \Sigma [(Benefits_t - Costs_t) \div (1 + r)^t]$. This simplified formula explains the concept of NPV; it is not a substitute for the Schedule IV-B Appendix A Cost Benefit Analysis (CBA) workbook.

### Discount Rate

For a Schedule IV-B, complete and submit the current Appendix A CBA workbook and use its prescribed calculations and fiscal-year cost-of-capital assumptions. For analyses outside Schedule IV-B, use an agency-approved financial policy and consistent timing and rate

conventions; obtain finance approval if no rate is prescribed. Any teaching or sensitivity-test rate (for example, 3%) must be labeled illustrative; it is not an FLDS or statewide standard.

| Apply NPV Consistently | Interpret the Result Carefully |
|---|---|
| 1. Use comparable alternatives. Include the current-state or do-minimum option, a common decision date, analysis horizon, and timing convention. | Positive NPV: the initiative creates net financial value compared with the baseline under the stated assumptions. |
| 2. Estimate incremental cash flows. Include implementation, staffing, integration, training, operations, licenses, transition, and decommissioning. For non-Schedule IV-B analysis, include residual value only when supported by agency financial policy; Schedule IV-B users should follow the workbook's prescribed categories. For portfolio decision analysis, do not let sunk costs drive the go-forward decision. For Schedule IV-B reporting, include current and previous project-related costs as required by the current Appendix A workbook. | Higher NPV: generally, favors one financially comparable discretionary project over another, but it does not establish the full public-sector priority. |
| 3. Separate financial and public value. Monetize only defensible cash impacts. Report service, compliance, safety, access, capacity, and risk reduction separately unless a validated monetary method exists. | Mandatory projects may have a negative NPV. Confirm the mandate and compare compliant alternatives to select the least-cost, best-value path. |
| 4. Use consistent assumptions. Apply the same rate, inflation basis, benefit timing, and cost classifications. Distinguish cash savings, cost avoidance, and released capacity. For Schedule IV-B, cost avoidance is treated as an intangible benefit and should not be entered in Appendix A as a tangible benefit. | Governance also considers strategic alignment, public value, security, compliance, urgency, dependencies, readiness, affordability, delivery risk, and available resources. |

| Apply NPV Consistently | Interpret the Result Carefully |
|---|---|
| 5. Test uncertainty and refresh. Run base, optimistic, and pessimistic cases. Stress schedule, adoption, volume, benefit timing, and discount rate; update the analysis when assumptions materially change. | Finance validates monetary assumptions; the business owner owns the benefits; the project manager maintains traceability and updates the analysis. |

Key principle: NPV is one decision input, not the sole approval test for a public-sector investment.

## A Balanced Portfolio Selection Method

Use the same documented process for every project so the decision is consistent, transparent, and auditable.

1. Gate mandatory projects. Identify statutory, regulatory, security, legislative, and critical operational work first and treat the mandate as a gating factor. Confirm the legal deadline, minimum compliant scope, consequences of noncompliance, available funding, and feasible alternatives. A mandate does not eliminate affordability, procurement, risk, or feasibility analysis.
2. Score discretionary projects. Use approved criteria and weights to compare strategic alignment, expected benefits and public value, risk reduction, customer impact, technical debt, delivery confidence, urgency, and organizational readiness.
3. Evaluate financial measures separately. For Schedule IV-B projects, use the Appendix A outputs: Return on Investment (ROI), Payback Period, Breakeven Fiscal Year, Net Present Value (NPV), and Internal Rate of Return (IRR). Benefit-Cost Ratio (BCR) may be used as a supplemental measure, but it is not an Appendix A output. Consider financial results alongside - not inside - the non-financial weighted score to avoid double-counting financial value.

Illustrative Non-Financial Weighted Scoring Model

| Criterion | Weight |
|---|---|
| Strategic alignment | 25% |
| Benefits and public value | 20% |
| Risk reduction | 15% |
| Urgency and organizational readiness | 15% |
| Customer or service impact | 10% |
| Technical debt reduction | 10% |
| Delivery confidence | 5% |

Weighted score = Σ [(rating ÷ 5) × weight]. Use the 1-5 evidence-based portfolio rating scale described in this guide. When weights are entered as percentage points (for example, 25 rather than 0.25), the maximum total score is 100. A rating of 4 on a 25% criterion contributes 20 points: (4 ÷ 5) × 25 = 20.

Important: This illustrative model intentionally excludes financial measures, so they are not counted twice. This scoring model is recommended portfolio-management guidance and is not prescribed by Florida statute, Rule Chapter 60GG-1, or Schedule IV-B. Agencies should tailor the non-financial criteria and weights to statutory responsibilities, strategic priorities, risk tolerance, and governance requirements.

## Applying the 1-5 Portfolio Rating Scale

The rating scale is intended to reduce guesswork and make scoring repeatable. Assign the rating that best matches the documented evidence for each criterion; do not score based on preference, advocacy, or an unsupported expectation of future performance.

Scale interpretation: These anchors assess the relative value and impact of proposed projects on a consistent 1-5 scale. They are intended for portfolio prioritization and should not be interpreted as an organizational maturity assessment.

| Rating | Anchor | Evidence-based guidance |
|---|---|---|
| 1 | Minimal | Little demonstrated value or alignment. Evidence is weak or missing; major assumptions or gaps remain. |
| 2 | Emerging | Some value or alignment is evident, but it is uneven or assumption dependent. Evidence is incomplete. |
| 3 | Moderate | Clear value or alignment is supported by reasonable evidence. Key assumptions are credible and gaps are manageable. |
| 4 | Significant | Strong value or alignment is supported by specific, credible evidence. Risks are understood and only minor gaps remain. |
| 5 | Transformational | Exceptional value or alignment is supported by compelling, well-validated evidence and measurable outcomes. |

### Rating Guidance

- Use criterion-specific evidence. Strategic alignment may rely on approved plans or mandates; benefits and public value should rely on measurable outcomes. For Schedule IV-B projects, focus the benefits/public value score on benefits not already monetized in Appendix A, or document how overlap is prevented. Risk reduction should rely on identified risks and expected exposure reduction; customer impact should rely on service data or user research; urgency, readiness, and delivery confidence should rely on documented deadlines, dependencies, capacity, and implementation evidence.
- Rate the current proposal, not the hoped-for future state. If evidence is not yet available, score conservatively and document what would be needed to justify a higher rating.
- Use the same anchors for every project in the same decision cycle. Do not change the meaning of a 3 or 4 from one project to another.

- Require a short evidence citation or rationale for every rating. Where reviewers disagree, discuss the evidence and record the final rationale rather than averaging unsupported opinions.
- Calibrate before formal scoring. Reviewers should jointly score one or two sample projects first, so they apply the anchors consistently.

## Using the Results to Make the Decision

Financial measures help explain value, while governance determines the best overall public investment.

| What Each Financial Measure Tells You | When Scores Are Close |
|---|---|
| • NPV: present-day value of project benefits less costs over the project lifecycle.<br>• Schedule IV-B ROI: dollar difference between Net Tangible Benefits and Project Cost.<br>• IRR: project rate of return; the discount rate at which NPV equals zero.<br>• Payback / Breakeven: time and fiscal year in which project costs are recovered.<br>• BCR (supplemental): present value of benefits divided by present value of costs; not an Appendix A output. | • Compare the strength of sensitivity results and the reliability of benefit assumptions.<br>• Consider feasibility: implementation risk, dependencies, resource capacity, procurement timing, and organizational readiness.<br>• Document the rationale. No single financial measure should determine the public-sector investment decision. |

### Illustrative Three-Project Decision

(Figures are examples only.)

| Initiative | Portfolio gate | NPV | ROI | Score | Decision context |
|---|---|---|---|---|---|
| Enterprise Resource Planning upgrade | Discretionary | $18M | 50% | 78 | Highest NPV; significant funding and resource demand. |

| Initiative | Portfolio gate | NPV | ROI | Score | Decision context |
|---|---|---|---|---|---|
| Cybersecurity remediation | Mandatory | $1M | 25% | Not scored | Advance required scope; select the least-cost compliant option. |
| Public portal | Discretionary | $8M | 267% | 86 | Highest discretionary portfolio score and IRR; validate adoption assumptions. |

Recommended approach: Identify mandatory work first and treat the mandate as a gating factor; assess the legal deadline, minimum compliant scope, consequences of noncompliance, available funding, and feasible alternatives. Rank discretionary projects with an approved non-financial weighted model, then combine the weighted score with Schedule IV-B financial outputs and feasibility constraints such as dependencies, affordability, and resource availability to support the final governance decision.

## How Do We Choose One Project When Funding Is Limited?

When multiple discretionary projects compete for limited funding, do not select a project based solely on NPV, IRR, or another single financial measure. Apply the same documented evaluation method to each alternative so the decision is consistent, transparent, repeatable, and auditable.

### Recommended Decision Rule

1. Identify mandatory projects first. Statutory, regulatory, security, legislative, or critical operational requirements should be addressed before discretionary projects.
2. Calculate financial value consistently. For Schedule IV-B projects, use the current Appendix A CBA workbook to calculate ROI, Payback Period, Breakeven Fiscal Year, NPV, and IRR using the workbook's prescribed assumptions. BCR may be used as a supplemental measure but is not an Appendix A output.
3. Score non-financial value. Evaluate strategic alignment, benefits and public value, risk reduction, customer/service impact, technical debt, delivery confidence, urgency, and organizational readiness using the approved scoring criteria and evidence.

4. Apply an approved non-financial weighted scoring model. Use the 1-5 evidence-based portfolio rating scale and predetermined weights. Require a documented rationale for each rating and retain the scoring record.
5. Use financial measures as separate decision inputs, not the sole decision rule. A project with the highest IRR may not create the greatest NPV, while a project with the highest NPV may not provide the greatest overall public value.
6. Document the decision. Record assumptions, Appendix A financial results, scoring criteria and weights, evidence supporting ratings, feasibility and risk considerations, resource constraints, approvals, and the rationale for selecting the project.

Bottom line: The decision question is not simply "Which project has the highest financial return?" It is "Which feasible project provides the greatest overall value to the organization and the public, based on consistently applied financial and non-financial criteria?"

### Minimum Decision Record

- Alternatives and baseline; cash-flow sources and classifications; current Appendix A CBA outputs (Schedule IV-B ROI, Payback Period, Breakeven Fiscal Year, NPV, and IRR); cost-of-capital assumptions; sensitivity results; and supplemental BCR if used.
- Scoring criteria, weights, 1-5 portfolio ratings, supporting evidence, reviewer/approval record, and any scoring calibration notes.
- Selection rationale, conditions, dependencies, resource constraints, and accountable benefit owner.

## Three-Project Example: If You Can Fund Only One

Use the same analysis for all three projects. The highest IRR does not automatically win; compare present-value financial results alongside the approved non-financial portfolio score and feasibility.

Example: Project A has a $10M NPV and 100% IRR; Project B has a $3M NPV and 150% IRR; Project C has a $10M NPV and 50% IRR. IRR alone would favor B, while NPV alone would show A and C as financially stronger in present-value dollars. The weighted non-financial score and governance

criteria help determine which project provides the best overall value. For Schedule IV-B projects, use the Appendix A workbook's calculated outputs.

### Key References

Section 216.023 and section 282.0051, Florida Statutes; Rule Chapter 60GG-1, Florida Administrative Code; Guidelines for Preparing the Schedule IV-B for Information Technology Projects; current Schedule IV-B Appendix A Cost Benefit Analysis workbook; Project Management Institute (PMI), A Guide to the Project Management Body of Knowledge (PMBOK® Guide), Eighth Edition; PMI Standard for Portfolio Management, Fourth Edition; and PMI Benefits Realization Management. Confirm current budget instructions and agency finance policy before use.